Practice Set B

Class 12 Mathematics, Chapter 11: Three Dimensional Geometry

Maximum marks: **51** **20** questions

Name _____ Date _____ Score _____ / 51

1. All questions are compulsory. Marks are shown against each question.
2. Show every step of working: steps carry marks, not just the final answer.
3. Where a question offers OR, attempt only one of the two choices.

Q No.	Question	Marks

Level 1

8 marks

1 MCQ, Board-pattern — 1 mark

The lines $\frac{x-1}{2}=\frac{y+1}{-1}=\frac{z}{3}$ and $\frac{x-3}{-4}=\frac{y+2}{2}=\frac{z-3}{-6}$ are:

(A) parallel but not coincident

(B) coincident

(C) intersecting at exactly one point

(D) skew

2 MCQ, NCERT Exemplar — 1 mark

A point A in space is such that $\overrightarrow{OA}$ is inclined at 60° to OX and at 45° to OY, and $|\overrightarrow{OA}|=10$ units. The position vector of A is:

(A) $5\hat{i}+5\sqrt{2}\hat{j}\pm 5\hat{k}$ **(B)** $5\sqrt{3}\hat{i}+5\sqrt{2}\hat{j}\pm 5\hat{k}$

(C) $5\hat{i}+5\sqrt{2}\hat{j}\pm 5\sqrt{2}\hat{k}$ **(D)** $\frac{1}{2}\hat{i}+\frac{1}{\sqrt{2}}\hat{j}\pm\frac{1}{2}\hat{k}$

3 MCQ, CBSE 2023 — 1 mark

Equation of a line passing through the point $(1,2,3)$ and equally inclined to the coordinate axes is:

(A) $\frac{x}{1}=\frac{y}{2}=\frac{z}{3}$ **(B)** $\frac{x-1}{1}=\frac{y-2}{2}=\frac{z-3}{3}$

(C) $\frac{x-1}{1}=\frac{y-1}{2}=\frac{z-1}{3}$ **(D)** $\frac{x-1}{1}=\frac{y-2}{1}=\frac{z-3}{1}$

4 MCQ, NCERT Misc. Ex. Q1 — 1 mark

The angle between the lines whose direction ratios are a, b, c and $b-c,\ c-a,\ a-b$ is:

(A) 0° **(B)** 30° **(C)** 60° **(D)** 90°

5 MCQ, Board-pattern — 1 mark

The lines $\frac{2x-1}{4}=\frac{1-y}{k}=\frac{z+2}{3}$ and $\frac{x-3}{k}=y+1=\frac{2-z}{2}$ are perpendicular to each other. The value of k is:

(A) 2 **(B)** -6 **(C)** 6 **(D)** -2

Q No.	Question	Marks
6	Assertion-Reason, CBSE SQP 2022-23 **Assertion (A):** The acute angle between the line $\vec{r} = \hat{i} + \hat{j} + 2\hat{k} + \lambda(\hat{i} - \hat{j})$ and the x-axis is $\frac{\pi}{4}$. **Reason (R):** The acute angle θ between the lines $\vec{r} = x_1\hat{i} + y_1\hat{j} + z_1\hat{k} + \lambda(a_1\hat{i} + b_1\hat{j} + c_1\hat{k})$ and $\vec{r} = x_2\hat{i} + y_2\hat{j} + z_2\hat{k} + \mu(a_2\hat{i} + b_2\hat{j} + c_2\hat{k})$ is given by $\cos\theta = \frac{\lvert a_1a_2 + b_1b_2 + c_1c_2\rvert}{\sqrt{a_1^2 + b_1^2 + c_1^2}\sqrt{a_2^2 + b_2^2 + c_2^2}}$. **(A)** Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A). **(B)** Both Assertion (A) and Reason (R) are true, but Reason (R) is **not** the correct explanation of the Assertion (A). **(C)** Assertion (A) is true, but Reason (R) is false. **(D)** Assertion (A) is false, but Reason (R) is true.	1 mark
7	Short answer, PYQ-style Find the vector and the Cartesian equations of the line passing through the points $A(0, 5, -2)$ and $B(3, -1, 2)$. Hence find the coordinates of the point on this line whose x-coordinate is 6.	2 marks

Level 2

16 marks

8	Short answer, NCERT Exemplar Prove that the lines $x = py + q,\ z = ry + s$ and $x = p'y + q',\ z = r'y + s'$ are perpendicular if $pp' + rr' + 1 = 0$.	2 marks
9	Short answer, CBSE 2025 Verify that the lines given by $\vec{r} = (1 - \lambda)\hat{i} + (\lambda - 2)\hat{j} + (3 - 2\lambda)\hat{k}$ and $\vec{r} = (\mu + 1)\hat{i} + (2\mu - 1)\hat{j} - (2\mu + 1)\hat{k}$ are skew lines. Hence, find the shortest distance between the lines.	3 marks
10	Short answer, Board-pattern Show that the lines $\frac{x-1}{2} = \frac{y+1}{-3} = \frac{z}{6}$ and $\frac{2x-6}{4} = -\frac{y}{3} = \frac{2z-2}{12}$ are parallel, and find the distance between them.	3 marks
11	Short answer, NCERT Exemplar Find the angle between the lines whose direction cosines are given by the equations $l + m + n = 0$ and $l^2 + m^2 - n^2 = 0$.	2 marks
12	Short answer, Board-pattern Find the value of a for which the lines $\vec{r} = (2\hat{i} - \hat{j}) + \lambda(a\hat{i} + 2\hat{j} - \hat{k})$ and $\frac{1-x}{3} = \frac{2y+1}{4} = z + 2$ are perpendicular.	2 marks
13	Short answer, NCERT Ex 11.1 Q5 Find the direction cosines of the sides of the triangle whose vertices are $A(3, 5, -4)$, $B(-1, 1, 2)$ and $C(-5, -5, -2)$.	2 marks

Q No.	Question	Marks
14	Short answer, Board-pattern Find the vector and the Cartesian equations of the line passing through the point $(2, -1, 5)$ and parallel to the line $4 - x = \frac{2y+1}{6} = \frac{3z-2}{-9}$.	2 marks

Level 3

27 marks

15	Long answer, NCERT Exemplar Find the foot of the perpendicular from the point $(2, 3, -8)$ to the line $\frac{4-x}{2} = \frac{y}{6} = \frac{1-z}{3}$. Also, find the perpendicular distance from the given point to the line.	5 marks
16	Long answer, PYQ-style Find the image P' of the point $P(2, -1, 5)$ in the line $\vec{r} = (11\hat{i} - 2\hat{j} - 8\hat{k}) + \lambda(10\hat{i} - 4\hat{j} - 11\hat{k})$. Also find the vector equation of the line joining P and P'.	5 marks
17	Long answer, CBSE 2023 Find the equations of the diagonals of the parallelogram $PQRS$ whose vertices are $P(4, 2, -6)$, $Q(5, -3, 1)$, $R(12, 4, 5)$ and $S(11, 9, -2)$. Use these equations to find the point of intersection of the diagonals.	5 marks
18	Long answer, NCERT Exemplar Find the equations of the two lines through the origin which intersect the line $\frac{x-3}{2} = \frac{y-3}{1} = \frac{z}{1}$ at angles of $\frac{\pi}{3}$ each.	5 marks
19	Short answer, CBSE SQP 2026-27 Find the angle between the lines $\frac{x-1}{2} = \frac{y-1}{2} = \frac{z-2}{1}$ and $\frac{x-5}{2} = y - \frac{1}{2} = \frac{3-z}{2}$. Also find the Cartesian equation of a line passing through the point $(1, 1, 1)$ and perpendicular to both lines.	3 marks
20	Case study, PYQ-style **Case Study:** Two insects crawl along straight paths in a garden. The first insect moves along the line $l_1 : \vec{r} = 6\hat{i} + 2\hat{j} + 2\hat{k} + \lambda(\hat{i} - 2\hat{j} + 2\hat{k})$ and the second along the line $l_2 : \vec{r} = -4\hat{i} - \hat{k} + \mu(3\hat{i} - 2\hat{j} - 2\hat{k})$. A biologist wants to know how close the two insects can ever get. Based on the given information, answer the following questions. **(i)** Write the direction ratios of the two paths and show that the paths are not parallel. — 1 mark **(ii)** Find a vector perpendicular to both the paths. — 1 mark **(iii)** Find the shortest distance between the two paths. — 2 marks	4 marks

Answer key

Chapter 11: Three Dimensional Geometry, Practice Set B. Full step-wise solutions are in the solutions sheet.

Level 1

1	(B) coincident
2	(A) $5\hat{i} + 5\sqrt{2}\hat{j} \pm 5\hat{k}$
3	(D) $\frac{x-1}{1} = \frac{y-2}{1} = \frac{z-3}{1}$
4	(D) 90°
5	(C) 6
6	(A) Both true; R is the correct explanation of A.
7	$\vec{r} = (5\hat{j} - 2\hat{k}) + \lambda(3\hat{i} - 6\hat{j} + 4\hat{k})$; $\frac{x}{3} = \frac{y-5}{-6} = \frac{z+2}{4}$; point $(6, -7, 6)$

Level 2

8	DRs are $p, 1, r$ and $p', 1, r'$; perpendicular when $pp' + 1 + rr' = 0$.
9	Skew lines; shortest distance $= \frac{8}{\sqrt{29}}$ units
10	Both have DRs $2, -3, 6$; distance $= \sqrt{5}$ units
11	$\frac{\pi}{3}$
12	$a = 1$
13	AB: $-\frac{2}{\sqrt{17}}, -\frac{2}{\sqrt{17}}, \frac{3}{\sqrt{17}}$; BC: $-\frac{2}{\sqrt{17}}, -\frac{3}{\sqrt{17}}, -\frac{2}{\sqrt{17}}$; CA: $\frac{4}{\sqrt{42}}, \frac{5}{\sqrt{42}}, -\frac{1}{\sqrt{42}}$
14	$\vec{r} = (2\hat{i} - \hat{j} + 5\hat{k}) + \lambda(\hat{i} - 3\hat{j} + 3\hat{k})$; $\frac{x-2}{1} = \frac{y+1}{-3} = \frac{z-5}{3}$

Level 3

15	Foot $(2, 6, -2)$; perpendicular distance $= 3\sqrt{5}$ units
16	Image $P'(0, 5, 1)$; line PP': $\vec{r} = (2\hat{i} - \hat{j} + 5\hat{k}) + \mu(\hat{i} - 3\hat{j} + 2\hat{k})$
17	PR: $\frac{x-4}{8} = \frac{y-2}{2} = \frac{z+6}{11}$; QS: $\frac{x-5}{2} = \frac{y+3}{4} = \frac{z-1}{-1}$; they meet at $\left(8, 3, -\frac{1}{2}\right)$
18	$\frac{x}{1} = \frac{y}{2} = \frac{z}{-1}$ and $\frac{x}{-1} = \frac{y}{1} = \frac{z}{-2}$
19	$\theta = \cos^{-1}\left(\frac{4}{9}\right)$; $\frac{x-1}{-5} = \frac{y-1}{6} = \frac{z-1}{-2}$
20	(i) DRs $1, -2, 2$ and $3, -2, -2$, not proportional; (ii) $8\hat{i} + 8\hat{j} + 4\hat{k}$; (iii) 9 units